

P2257

Diagnostic information - DME (DFI) control unit

Secondary-air pump relay driver

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in lines for secondary-air pump relay activation
- ♦ Secondary-air pump relay faulty

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Diagnostic information - DME (DFI) control unit

Secondary-air pump relay driver

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in lines for secondary-air pump relay activation
- ♦ Secondary-air pump relay faulty

Diagnostic information - DME (DFI) control unit

Boost pressure too low for tank ventilation

Diagnostic conditions

- Engine coolant temperature > 60 °C (140 °F)
- 3 full-load accelerations:
 - 3rd gear (no gearshift)
 - Engine speed from 1,500 to 5,000 rpm

Possible fault causes

- ♦ Insufficient vacuum supply due to leak in vacuum system or faulty vacuum pump (► see ⓘ 2)
- ♦ Positive crankcase ventilation check valve leaking
- ♦ Leak between turbocharger (on pressure side) and intake valve (engine)
- ♦ Vacuum hose leaking at boost pressure control cycle valve
- ♦ Diverter solenoid valve not installed or installed incorrectly
- ♦ Diverter solenoid valve leaking or stuck
- ♦ Diverter valve not installed or installed incorrectly
- ♦ Diverter valve leaking or stuck open (► see ⓘ 1)
- ♦ Boost pressure control cycle valve stuck open electrically or mechanically
- ♦ Wastegate mechanical fault, leaking or stuck open
- ♦ Turbocharger (1 or 2) sluggish or stuck

Fault setting condition(s)

- Measured boost pressure < required boost pressure to operate tank ventilation via turbocharger (map over load and elevation)

Fault effects

- If consequential fault of P0299, then:
 - ♦ Boost pressure control switched off (boost pressure low)
 - ♦ Engine torque is reduced

ⓘ INFORMATION 1

When the diverter valves are in normal position, they are closed by a spring, but can be pressed on by boost pressure.

They each have two connections:

- ♦ a boost pressure connection for closing the valve and keeping it closed (branch ahead of throttle valve);
- ♦ a vacuum connection for opening the valve; switched by a solenoid valve.

ⓘ INFORMATION 2

The vacuum system includes:

- ♦ Mechanical vacuum pump
- ♦ Vacuum check valve
- ♦ Brake booster
- ♦ Coolant shutoff valve and relevant solenoid change-over valve, respectively for:
 - ♦ Heat exchanger for heating

- ♦ Engine (short-circuit loop)
- ♦ Heat exchanger for transmission oil
- ♦ Heat exchanger for clutch fluid (PDK only)
- ♦ Vacuum unit and relevant solenoid change-over valve, respectively for:
 - ♦ Exhaust flaps
 - ♦ Acoustic simulators
 - ♦ Diverter valve s
 - ♦ Wastegates
- ♦ Vacuum lines to all components mentioned above

Diagnostic information - DME (DFI) control unit

Exhaust gas implausible, bank 1

Diagnostic conditions

Perform all short tests in sequence (without tank leak test)

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

- ♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty
- ♦ Catalytic converter, bank 1 faulty

Fault setting condition(s)

- The oxygen sensor upstream of catalytic converter, bank 1 measures the corresponding oxygen value after mixture lean-out/enrichment.
- Despite a sufficiently long waiting time, the corresponding voltage limit value (< 150 mV / > 800 mV) is not reached at the oxygen sensor downstream of catalytic converter, bank 1.

Diagnostic information - DME (DFI) control unit

Exhaust gas implausible, bank 1

Diagnostic conditions

Perform all short tests in sequence (without tank leak test)

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
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- Lambda control ahead of catalytic converter active
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- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

- ♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty
- ♦ Catalytic converter, bank 1 faulty

Fault setting condition(s)

- The oxygen sensor upstream of catalytic converter, bank 1 measures the corresponding oxygen value after mixture lean-out/enrichment.
- Despite a sufficiently long waiting time, the corresponding voltage limit value (< 150 mV / > 800 mV) is not reached at the oxygen sensor downstream of catalytic converter, bank 1.

Diagnostic information - DME (DFI) control unit

Exhaust gas implausible, bank 2

Diagnostic conditions

Perform all short tests in sequence (without tank leak test)

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty
- ♦ Catalytic converter, bank 1 faulty

Fault setting condition(s)

- The oxygen sensor upstream of catalytic converter, bank 2 measures the corresponding oxygen value after mixture lean-out/enrichment.
- Despite a sufficiently long waiting time, the corresponding voltage limit value (< 150 mV / > 800 mV) is not reached at the oxygen sensor downstream of catalytic converter, bank 2.

Diagnostic information - DME (DFI) control unit

Exhaust gas implausible, bank 2

Diagnostic conditions

Perform all short tests in sequence (without tank leak test)

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty
- ♦ Catalytic converter, bank 1 faulty

Fault setting condition(s)

- The oxygen sensor upstream of catalytic converter, bank 2 measures the corresponding oxygen value after mixture lean-out/enrichment.
- Despite a sufficiently long waiting time, the corresponding voltage limit value (< 150 mV / > 800 mV) is not reached at the oxygen sensor downstream of catalytic converter, bank 2.

Diagnostic information - DME (DFI) control unit

Fuel high pressure adaptation

INFORMATION 1

- ♦ Faults in the fuel high-pressure injection system area are frequently caused by defects in the pre-delivery circuit (fuel low pressure).
- ♦ Therefore, the pre-delivery circuit must be checked first in case of problems in the area of the fuel high-pressure injection system.

Diagnostic conditions

- Engine speed 450 ... 4,000 rpm
- Engine temperature - 10 ... + 100 °C (14 ... 212 °F)
- Driving with different operating conditions

Possible fault causes

- ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
- ♦ Fuel high-pressure adaptation required (see  2)
- ♦ Fuel high-pressure pump faulty

INFORMATION 2

The old values must be deleted prior to a new adaptation of the fuel high-pressure ranges.

- ♦ The fuel high-pressure adaptation values can be deleted using the **PIWIS Tester** as follows:

DME ⇒ Maintenance/repairs ⇒ Delete adaptation values

- ♦ The new adaptation takes place while driving.

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Diagnostic information - DME (DFI) control unit

Quantity control valve output stage

Diagnostic conditions

- Ignition on > 5 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Quantity control valve faulty (in fuel high-pressure pump)

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Diagnostic information - DME (DFI) control unit

Quantity control valve output stage

Diagnostic conditions

- Ignition on > 5 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Quantity control valve faulty (in fuel high-pressure pump)

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Diagnostic information - DME (DFI) control unit

Quantity control valve output stage

Diagnostic conditions

- Ignition on > 5 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Quantity control valve faulty (in fuel high-pressure pump)

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1 - implausible signal (during overrun fuel cut-off)

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Driving with:
 - Exhaust temperature > 200 °C (392 °F)
 - Overrun phase(s) > 8 second(s)

Possible fault causes

- ♦ Unintentional fuel supply, e.g. due to:
 - ♦ dripping fuel injector
 - ♦ high fuel content in engine oil
 - ♦ large quantity of evaporating engine oil
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Oxygen sensor upstream of catalytic converter, bank 1 faulty (aged)

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1 - implausible signal (during overrun fuel cut-off)

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
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or

- Driving with:
 - Exhaust temperature > 200 °C (392 °F)
 - Overrun phase(s) > 8 second(s)

Possible fault causes

- ♦ Unintentional fuel supply, e.g. due to:
 - ♦ dripping fuel injector
 - ♦ high fuel content in engine oil
 - ♦ large quantity of evaporating engine oil
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Oxygen sensor upstream of catalytic converter, bank 2 faulty (aged)

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Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 1

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 1 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 1

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

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- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 1 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 2

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 2 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 2

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 2 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 3

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 3 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 3

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 3 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 4

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 4 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 4

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 4 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 5

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 5 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 5

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 5 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 6

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 6 faulty

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Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 6

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 6 faulty

P2400

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module pump driver

Diagnostic conditions

- Run [DMTL pump](#) drive link

or

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature ([T020](#)) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature ([T052](#)) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure ([A060](#)) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start ([T072](#)) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ Open circuit, short circuit to B+ or short circuit to ground in activation line to pump motor for DMTL module
- ♦ Open circuit in the power supply to the DMTL module
- ♦ DMTL - tank leakage diagnosis module faulty

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module pump driver

Diagnostic conditions

- Run [DMTL pump](#) drive link

or

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature ([T020](#)) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature ([T052](#)) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure ([A060](#)) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start ([T072](#)) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ Open circuit, short circuit to B+ or short circuit to ground in activation line to pump motor for DMTL module
- ♦ Open circuit in the power supply to the DMTL module
- ♦ DMTL - tank leakage diagnosis module faulty

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module pump driver

Diagnostic conditions

- Run [DMTL pump](#) drive link

or

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature ([T020](#)) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature ([T052](#)) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure ([A060](#)) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start ([T072](#)) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ Open circuit, short circuit to B+ or short circuit to ground in activation line to pump motor for DMTL module
- ♦ Open circuit in the power supply to the DMTL module
- ♦ DMTL - tank leakage diagnosis module faulty

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module

Diagnostic conditions

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature (T020) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature (T052) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure (A060) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start (T072) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ DMTL - tank leakage diagnosis module faulty

P240A

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module heater driver

Diagnostic conditions

- Run [DMTL heater](#) drive link

or

- Engine running

Possible fault causes

- ♦ Open circuit, short circuit to B+ or short circuit to ground in heater activation line for DMTL module
- ♦ Open circuit in the power supply to the DMTL module
- ♦ DMTL - tank leakage diagnosis module faulty

P240B

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module heater driver

Diagnostic conditions

- Run [DMTL heater](#) drive link

or

- Engine running

Possible fault causes

- ♦ Open circuit, short circuit to B+ or short circuit to ground in heater activation line for DMTL module
- ♦ Open circuit in the power supply to the DMTL module
- ♦ DMTL - tank leakage diagnosis module faulty

P240C

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module heater driver

Diagnostic conditions

- Run [DMTL heater](#) drive link

or

- Engine running

Possible fault causes

- ♦ Open circuit, short circuit to B+ or short circuit to ground in heater activation line for DMTL module
- ♦ Open circuit in the power supply to the DMTL module
- ♦ DMTL - tank leakage diagnosis module faulty

P2414

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1 - sensor in air

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

• Perform all short tests in sequence (without tank leak test).

or

• Lambda control active

• Driving > 10 second(s) with:

• No overrun phase(s)

• No fault entered for: oxygen sensor(s) or mixture preparation

Possible fault causes

♦ Oxygen sensor upstream of catalytic converter, bank 1 not installed (only electrically connected)

♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

♦ Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see  below)

INFORMATION

Example

♦ Vehicle speed approx. 140 km/h (86 mph)

♦ Little or no acceleration although accelerator pedal is floored

♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:

♦ Fault entry P2414

♦ Oxygen sensing, bank 1 = 25.0% (enrichment)

♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%

♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V

♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

i The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure increase that is too slow, or a rapid drop in pressure when the fuel injectors open.
- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

P2415

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2 - sensor in air

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Lambda control active
- Driving > 10 second(s) with:
 - No overrun phase(s)
- No fault entered for: oxygen sensor(s) or mixture preparation

Possible fault causes

- ♦ Oxygen sensor upstream of catalytic converter, bank 2 not installed (only electrically connected)
- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty
- ♦ Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see  below)

INFORMATION

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
 - ♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%
 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

i The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure increase that is too slow, or a rapid drop in pressure when the fuel injectors open.
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Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

P2450

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module

Diagnostic conditions

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature (T020) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature (T052) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure (A060) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start (T072) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ DMTL - tank leakage diagnosis module faulty

P24D6

Diagnostic information - DME (DFI) control unit

Tank vent pressure sensor: electrical fault

Diagnostic conditions

- Ignition on or engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Ground supply faulty
- ♦ Tank vent pressure sensor faulty
- For fault code P24D6:
 - ♦ Check valve between tank vent valve and intake manifold leaking or installed backward

P24D7

Diagnostic information - DME (DFI) control unit

Tank vent pressure sensor: electrical fault

Diagnostic conditions

- Ignition on or engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Ground supply faulty
- ♦ Tank vent pressure sensor faulty
- For fault code P24D6:
 - ♦ Check valve between tank vent valve and intake manifold leaking or installed backward